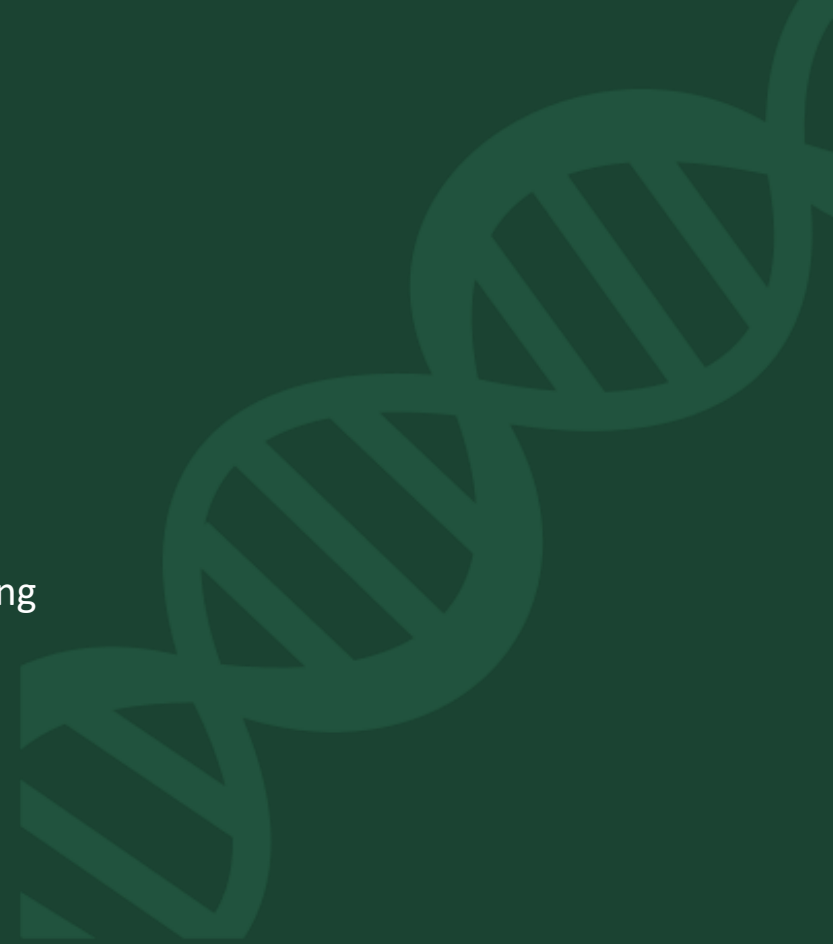


HYBRIDS · LEGISLATION · MANAGEMENT

Hybrids in conservation: threat, resource, or both?

Based on Galbusera et al. (2025) Hybrids along a natural–anthropogenic gradient: improving policy and management across all levels of biodiversity.

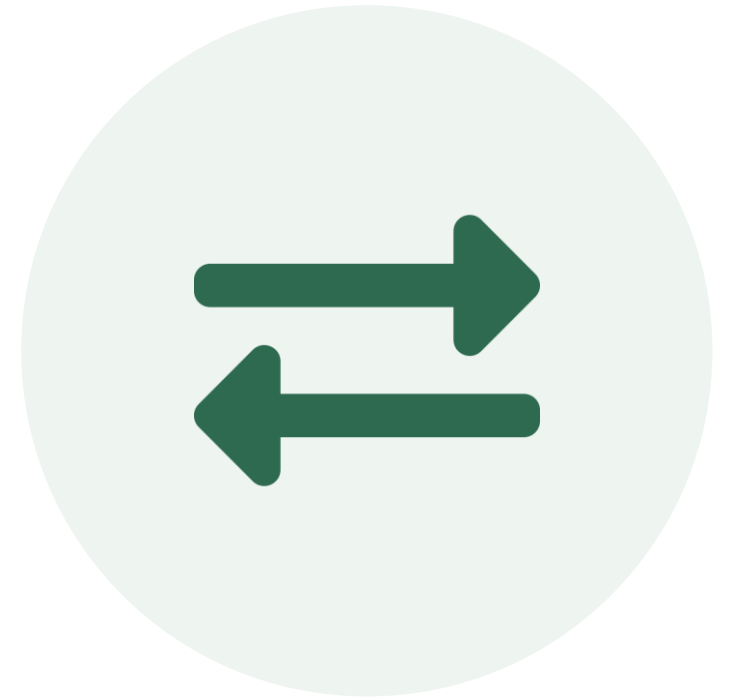
Conservation Letters 18(6): e13158

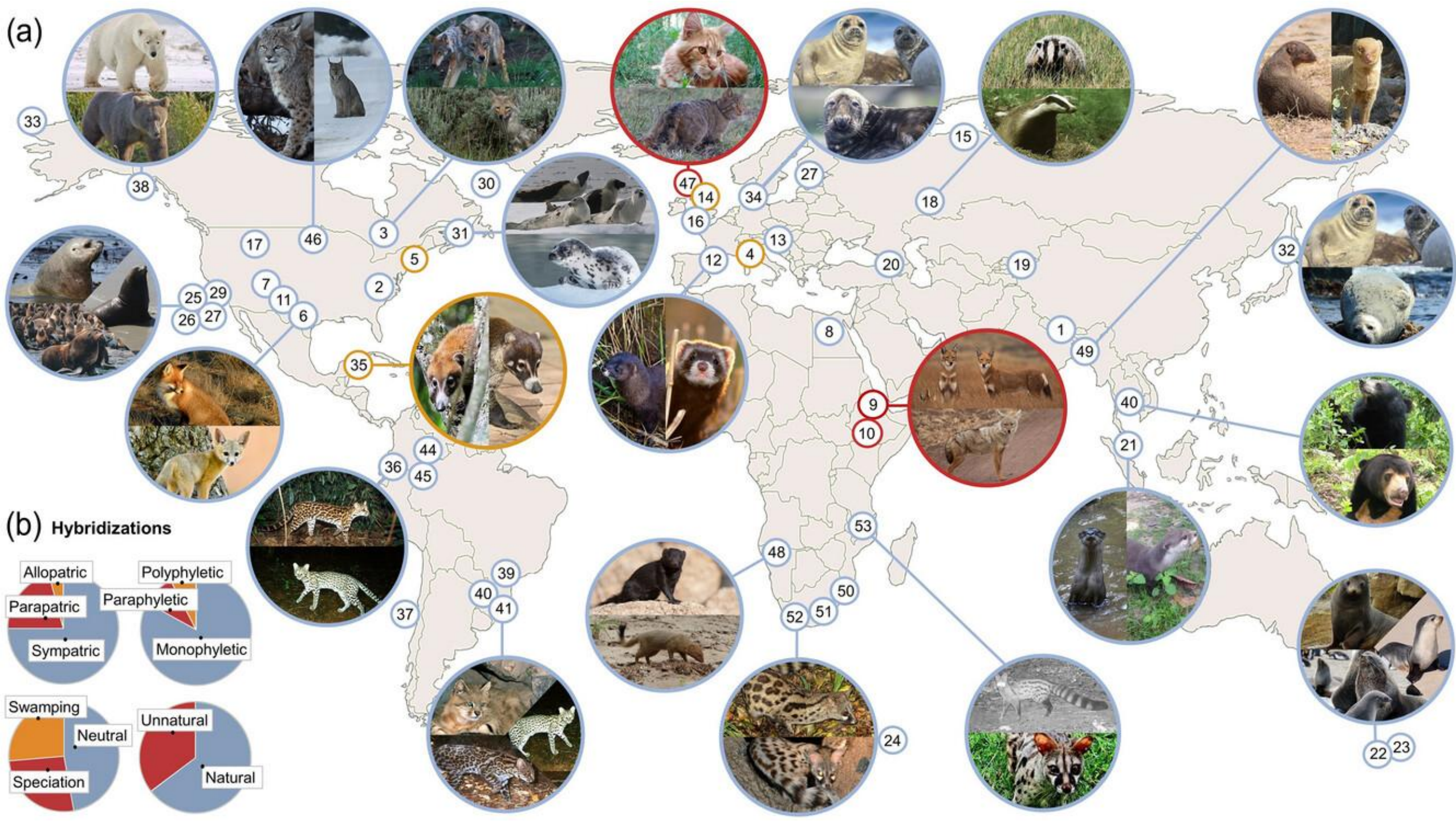


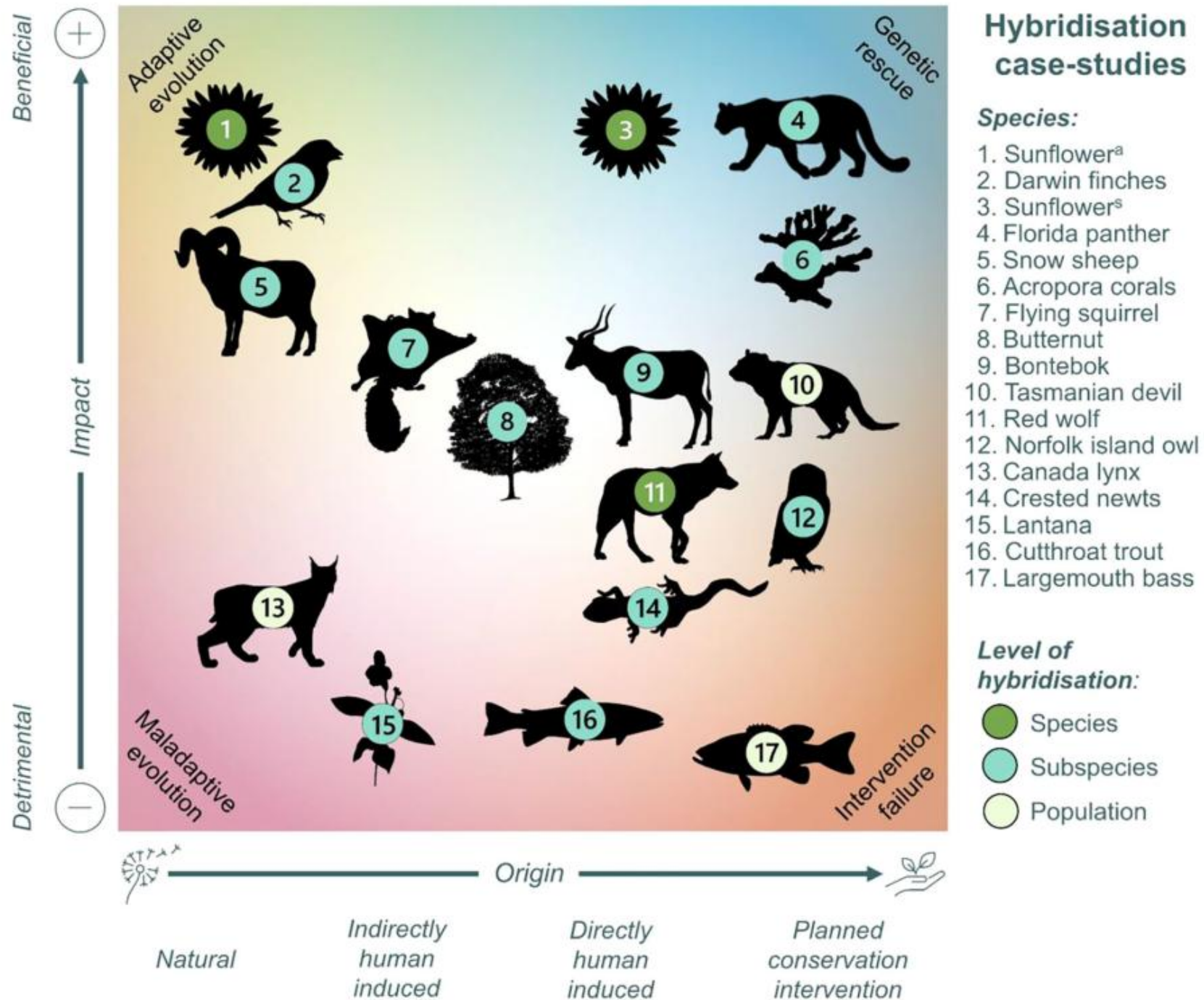
1. The ambiguity of “hybrids”

Core terms

- **Hybridization:** mating between different species, subspecies, or genetically distinct populations.
- **Introgression:** hybrids backcross with parental groups, moving genes across lineages.
- Hybridization can be **natural** (evolutionary process) **or connected to human actions** (directly or indirectly).



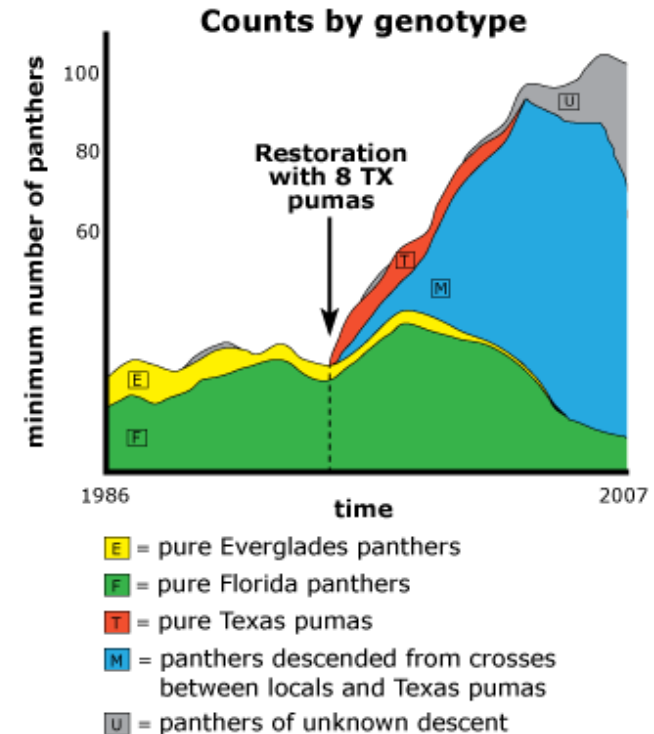
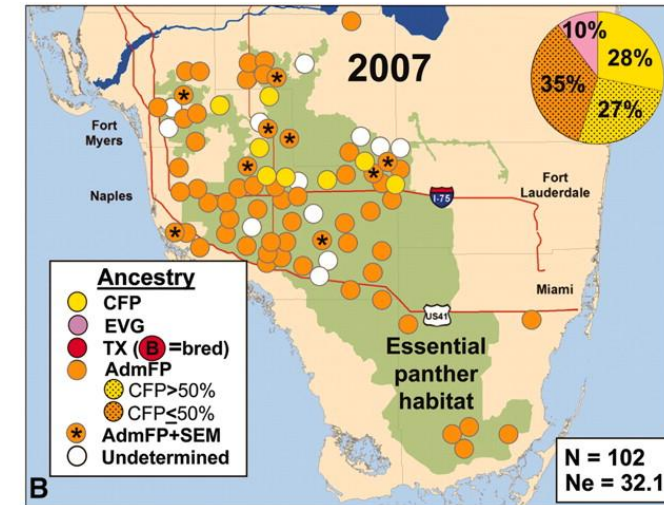
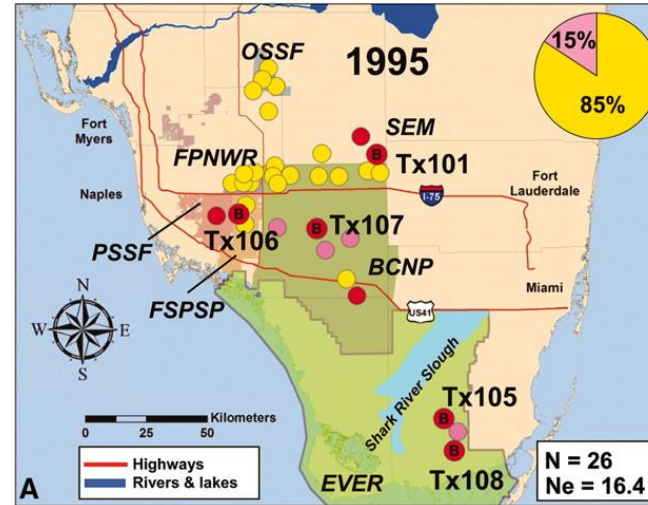




Genetic Restoration of the Florida Panther

Warren E. Johnson,^{1*†} David P. Onorato,^{2*†} Melody E. Roelke,^{3*} E. Darrell Land,^{2*} Mark Cunningham,² Robert C. Belden,⁴ Roy McBride,⁵ Deborah Jansen,⁶ Mark Lotz,² David Shindle,² JoGayle Howard,⁸ David E. Wildt,⁸ Linda M. Penfold,⁹ Jeffrey A. Hostetler,¹⁰ Madan K. Oli,¹⁰ Stephen J. O'Brien^{1†}

The rediscovery of remnant Florida panthers (*Puma concolor coryi*) in southern Florida swamplands prompted a program to protect and stabilize the population. In 1995, conservation managers translocated eight female pumas (*P. c. stanleyana*) from Texas to increase depleted genetic diversity, improve population numbers, and reverse indications of inbreeding depression. We have assessed the demographic, population-genetic, and biomedical consequences of this restoration experiment and show that panther numbers increased threefold, genetic heterozygosity doubled, survival and fitness measures improved, and inbreeding correlates declined significantly. Although these results are encouraging, continued habitat loss, persistent inbreeding, infectious agents, and possible habitat saturation pose new dilemmas. This intensive management program illustrates the challenges of maintaining populations of large predators worldwide.





Genetic rescue of Florida panthers reduced homozygosity but did not swamp ancestral genotypes

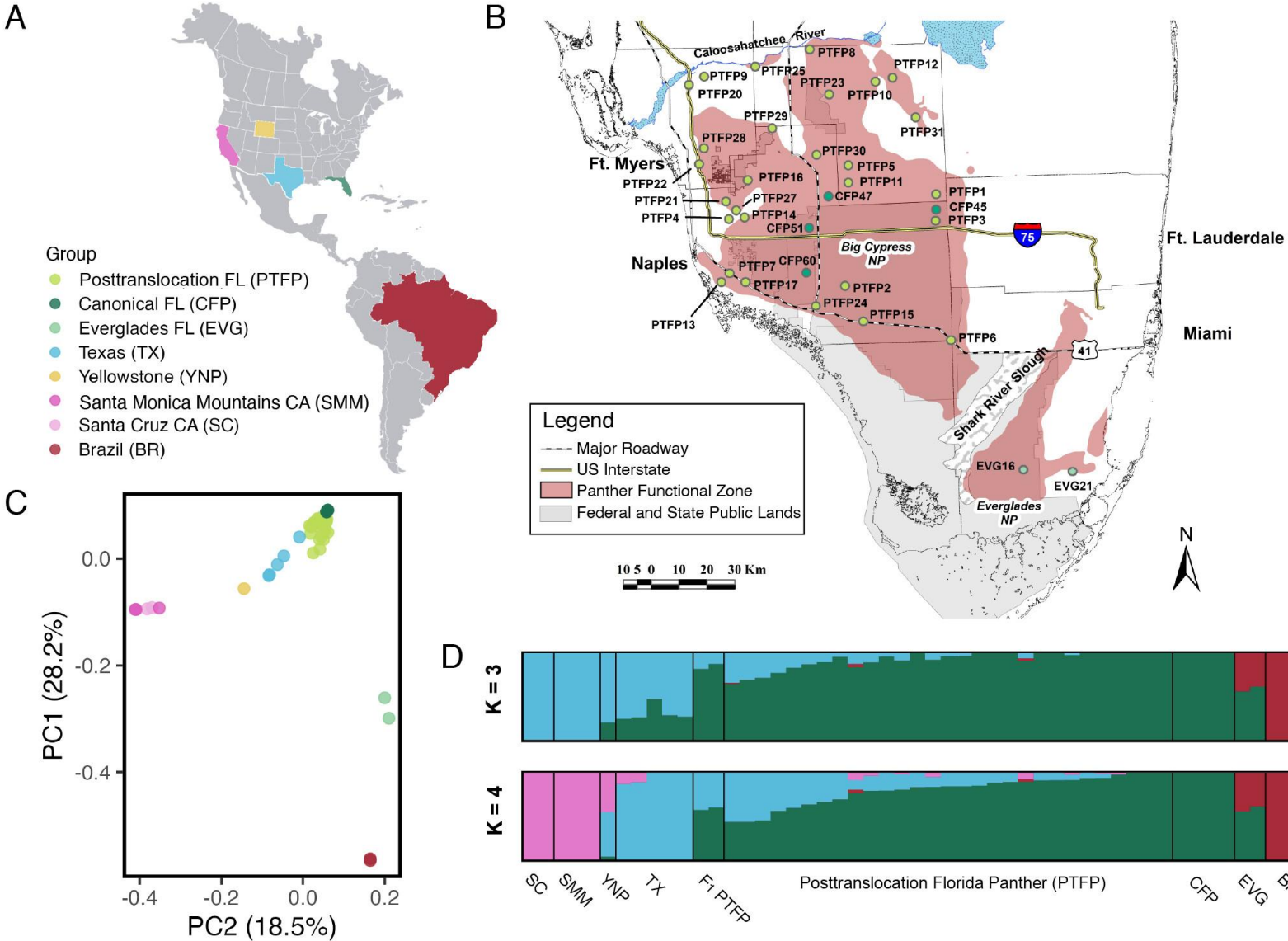
Diana Aguilar-Gómez^{a,b,1} , Lin Yuan^c , Yulin Zhang^b, Alexander Ochoa^d, Melanie Culver^{a,f}, Robert R. Fitak^e , Dave Onorato^h , Kirk E. Lohmueller^{a,i,1} , and Rasmus Nielsen^{b,1}

Affiliations are included on p. 11.

Edited by Anne Yoder, Duke University, Durham, NC; received June 28, 2024; accepted February 7, 2025

Significance

In the mid 1990s, Florida panthers were at risk of extinction due to isolation and habitat loss with the population numbering less than 30 individuals, many of whom exhibited morphologic and genetic correlates of inbreeding. In 1995, eight female pumas from Texas were translocated to Florida to attempt “genetic rescue.” We show that the rescue was successful with observed improvements due to increased heterozygosity rather than reduction in the number of deleterious variants. Further, our analysis shows that the Florida genetic ancestry was not completely replaced, thus allaying fears that rescue leads to extinction by replacement. We demonstrate that knowledge gained from speciation science can be applied to conservation action via the mechanism of genetic rescue from appropriately diverged populations.



Increased Infectious Disease Susceptibility Resulting from Outbreeding Depression

TONY L. GOLDBERG,*†‡§ EMILY C. GRANT,† KATE R. INENDINO,† TODD W. KASSLER,**
JULIE E. CLAUSSEN,‡ AND DAVID P. PHILIPP†‡§

*Department of Veterinary Pathobiology, University of Illinois at Urbana-Champaign, 2001 South Lincoln Avenue, Urbana, IL 61802, U.S.A., email tlgoldbc@uiuc.edu

†Department of Natural Resources and Environmental Sciences, University of Illinois at Urbana-Champaign, 1102 South Goodwin Avenue, Urbana, IL 61801, U.S.A.

‡Illinois Natural History Survey, Center for Aquatic Ecology, 607 East Peabody Drive Champaign, IL 61820, U.S.A.

§Program in Ecology and Evolutionary Biology, University of Illinois at Urbana-Champaign, 286 Morrill Hall, 505 South Goodwin Avenue, Urbana, IL 61801, U.S.A.

**Washington Department of Fish and Wildlife, 600 Capitol Way North, Olympia, WA 98501-1091, U.S.A.

Can outbreeding depression reduce fitness by making hybrids more susceptible to infectious disease?

- They studied the responses of largemouth bass (*Micropterus salmoides*) to a newly discovered viral pathogen of North American warmwater fishes, largemouth bass virus
- F2 generation hybrids suffered mortality at a rate 3.6 times higher than either F1 generation hybrids or wild-type parental fish
- Outbreeding may disrupt coadapted immune gene complexes, especially in F2 generations.

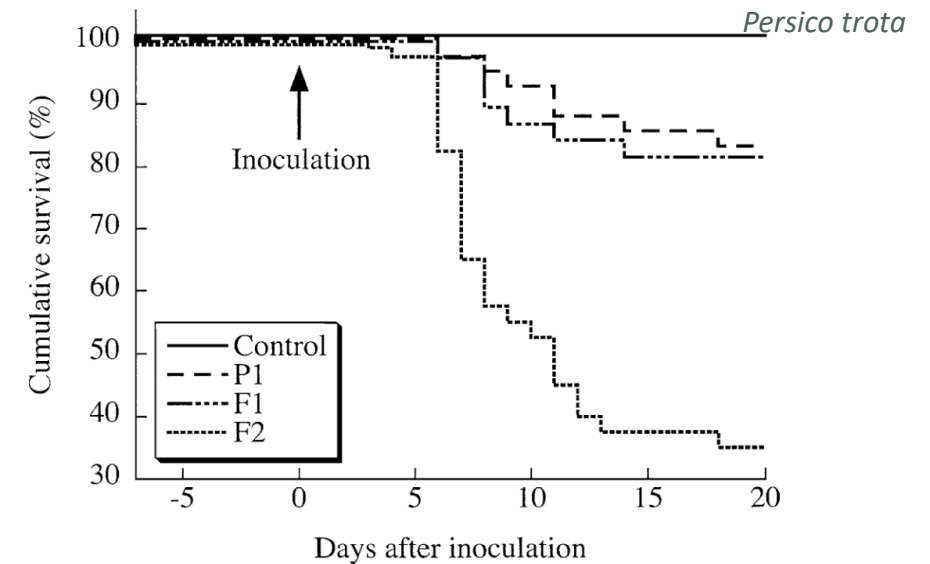
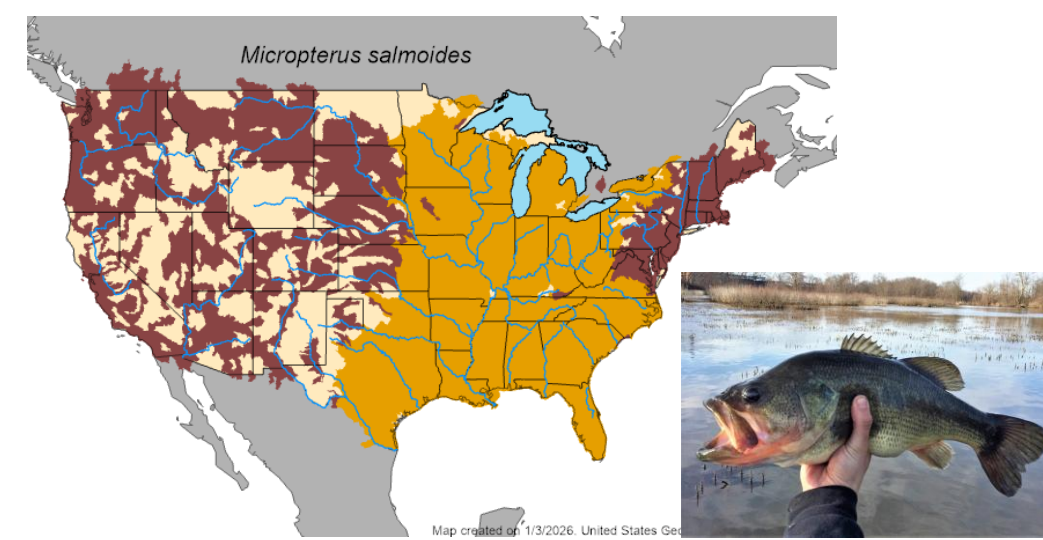


Figure 2. Cumulative survival of largemouth bass exposed to largemouth bass virus. No mortality was observed in sham-injected control fish of any level of outbreeding (P1 parental stocks, F1 generation outcrosses, or F2 generation outcrosses), so results from all control fish were combined and represented as a single line.

1. The ambiguity of “hybrids”

Genomic data and new methods

- Improved **detection** of hybrids.
- Hybrids occurring between a greater number of taxa than previously thought.
- Hybrids were found in about 23% of all carnivore species.
- In the future, **anthropogenic disturbance** might increase the rates of hybridization.
- Environmental policy makers and natural resource managers are increasingly challenged to **handle and manage hybrids**.



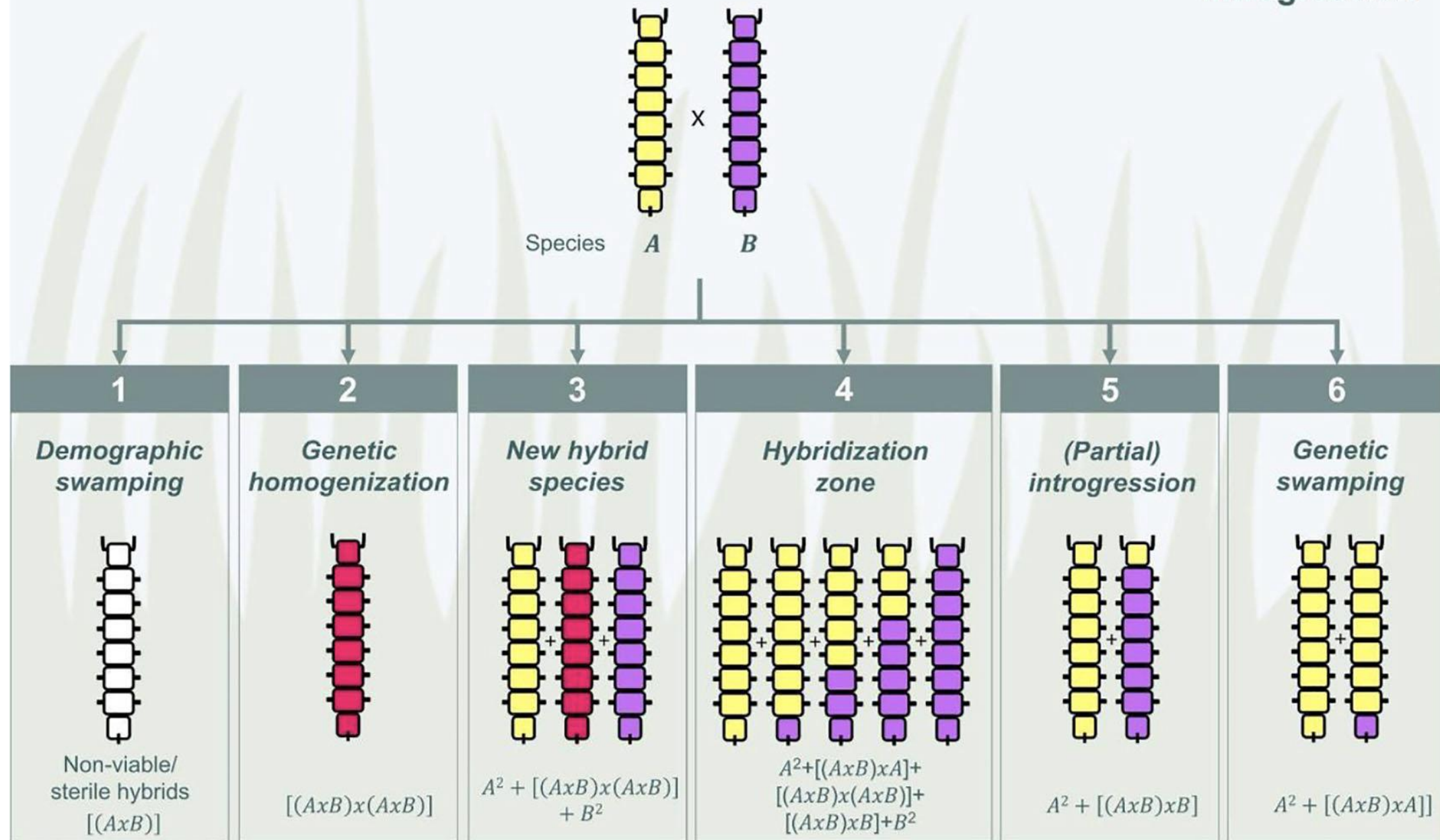
1. The ambiguity of “hybrids”

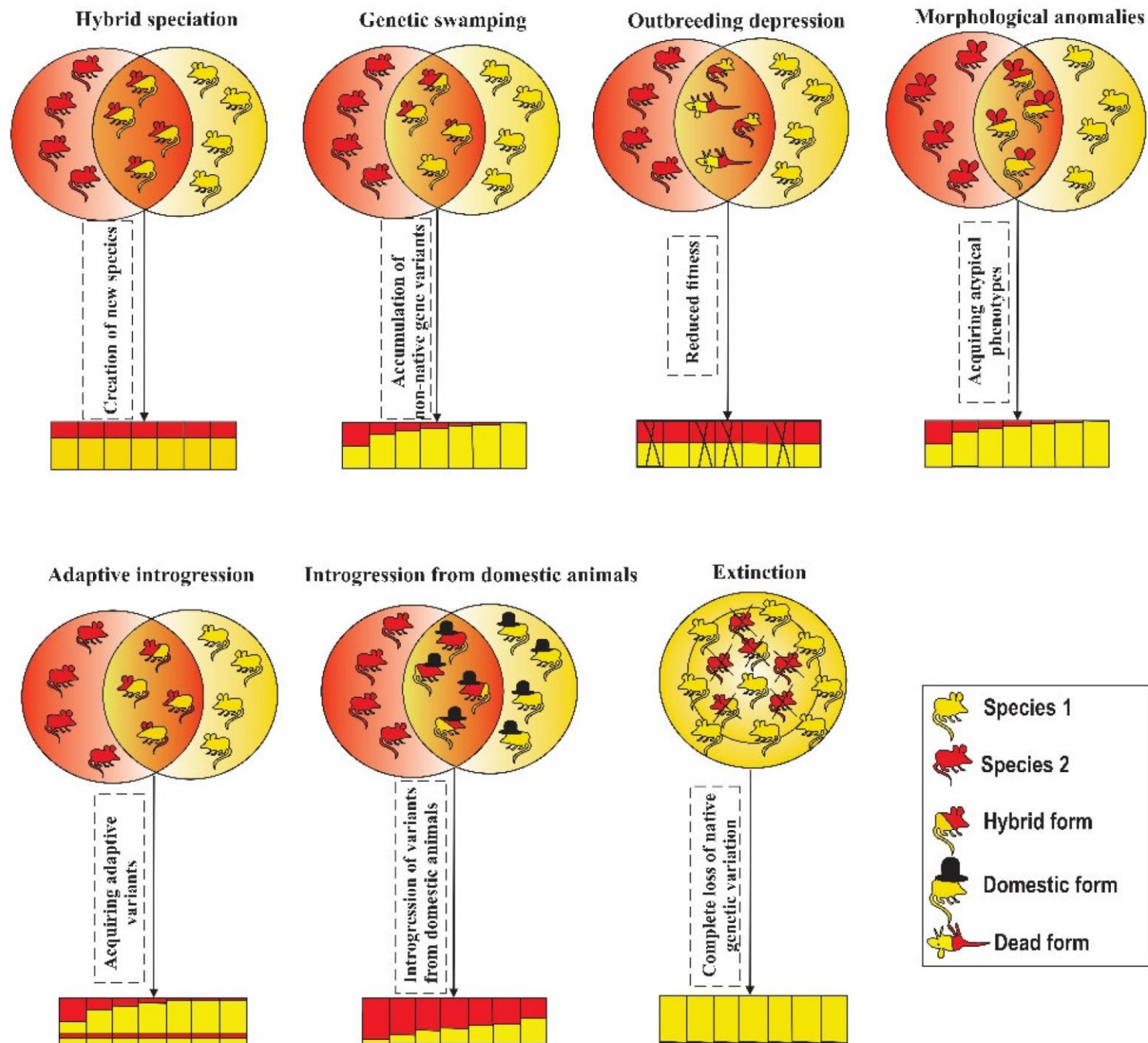
Negative aspects

- In early conservation literature, hybrids between native and non-native taxa were primarily considered a **threat**.
- Such hybrids could lead to loss of biodiversity by replacing resident taxa through **introgression and genetic swamping**.
- Specific alleles that underlie **local adaptation** may **disappear**, or hybrids may introduce **maladaptive alleles** given local climates or pathogens.
- Hybrids can also reduce fitness via reduced fertility and **outbreeding depression**.



Hybridisation and introgression





1. The ambiguity of “hybrids”

Negative aspects

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- Such hybrids could lead to loss of biodiversity by replacing resident taxa through introgression and genetic swamping.
- Specific alleles that underlie local adaptation may disappear, or hybrids may introduce maladaptive alleles given local climates or pathogens.
- Hybrids can also reduce fitness via reduced fertility and outbreeding depression.

**At the species level,
few extinctions
caused by hybrids
have been
documented.**

1. The ambiguity of “hybrids”

Positive aspects

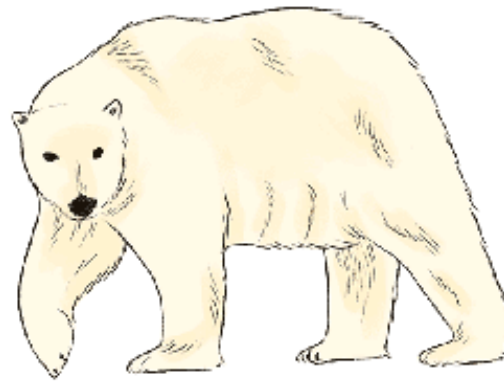
- For taxa at the brink of extinction and displaying negative growth rates, mixing of different gene pools can improve and **restore genetic variation** (heterosis or “hybrid vigor”).
- Hybrids might harbor genetic diversity of species that are extinct, extirpated, or at risk.
- Hybrids may also fulfill an important role in an ecosystem, maintaining certain processes (e.g., predator–prey relationships).



Identified knowledge gaps in hybridization research

Selective pressures on hybrids under rapid environmental change

Example: hybridization among bears (*Ursus* spp.); fitness effects of introgressed alleles can be hard to measure and may shift over time and context.



A. Preuß et al. · Bear-hybrids: behaviour and phenotype

209



Fig. 4. Comparison between the frames of the female hybrids (A) and the male hybrid (B) to a polar bear (C) and a brown bear (D).

Identified knowledge gaps in hybridization research

Behavioral ecology of hybrids in human-dominated landscapes

Human-altered habitats and anthropogenic food sources may impose new selective forces compared with historical conditions of parental species.

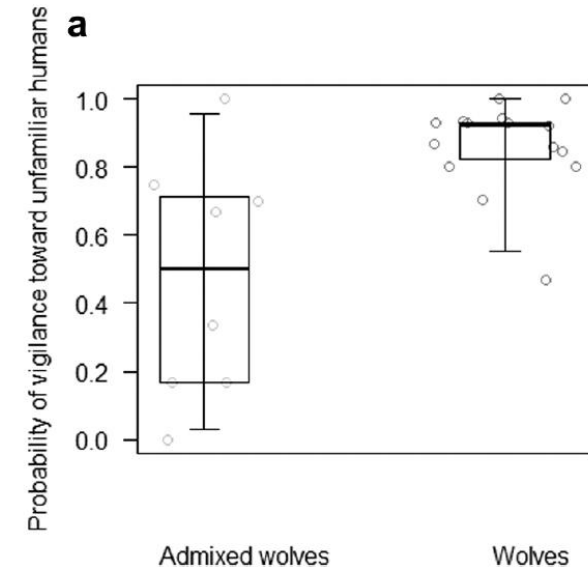
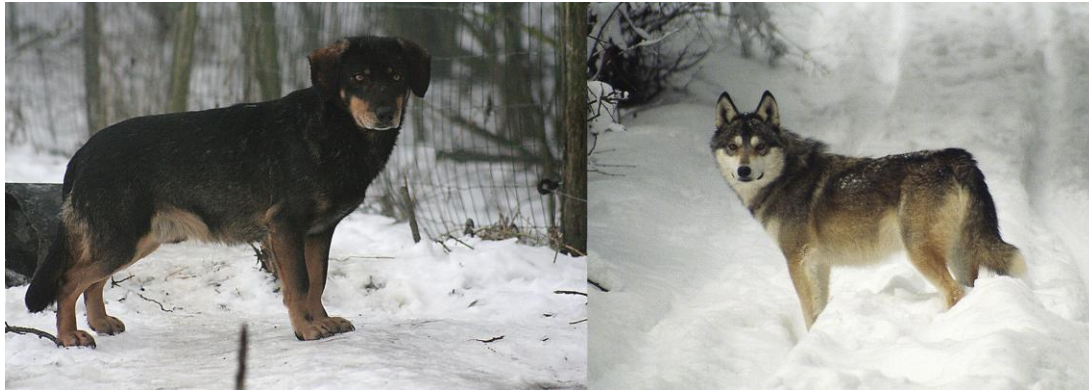
Animal Cognition (2024) 27:9
<https://doi.org/10.1007/s10071-024-01849-7>

ORIGINAL PAPER



A first exploratory comparison of the behaviour of wolves (*Canis lupus*) and wolf-dog hybrids in captivity

Federica Amici^{1,2} · Simone Meacci³ · Emmeline Caray⁴ · Linda Oña¹ · Katja Liebal^{1,2} · Paolo Ciucci³



When exposed to unfamiliar humans, admixed individuals were less vigilant, less fearful and less aggressive than wolves, and they were also more likely to approach humans, without showing fearful or aggressive behaviour.

Identified knowledge gaps in hybridization research

Better characterization of heterogeneous genomic outcomes of hybridization

Need to map which genes introgress, especially “barrier loci” linked to reproductive isolation, and understand gene-centric histories that may not match species phylogenies.

How genetic differentiation between diverging lineages is heterogeneously distributed across the genome, often concentrated in a few genomic regions, so-called “islands of differentiation”.

The genomic islands may harbor loci that contribute to reproductive isolation, and these so-called barrier loci are thus less likely to introgress compared to neutral loci.

STUDY DESIGNS

Making sense of genomic islands of differentiation in light of speciation

Jochen B. W. Wolf^{1,2} and Hans Ellegren¹

Abstract | As populations diverge, genetic differences accumulate across the genome. Spurred by rapid developments in sequencing technology, genome-wide population surveys of natural populations promise insights into the evolutionary processes and the genetic basis underlying speciation. Although genomic regions of elevated differentiation are the focus of searches for ‘speciation genes’, there is an increasing realization that such genomic signatures can also arise by alternative processes that are not related to population divergence, such as linked selection. In this Review, we explore methodological trends in speciation genomic studies, highlight the difficulty in separating processes related to speciation from those emerging from genome-wide properties that are not related to reproductive isolation, and provide a set of suggestions for future work in this area.

Identified knowledge gaps in hybridization research

How species boundaries are preserved despite horizontal genetic exchange

Priority: identify tipping points where introgression leads to genetic swamping, and drivers of future boundary shifts to improve risk management.

Kraus et al. *BMC Evolutionary Biology* 2012, **12**:45
<http://www.biomedcentral.com/1471-2148/12/45>

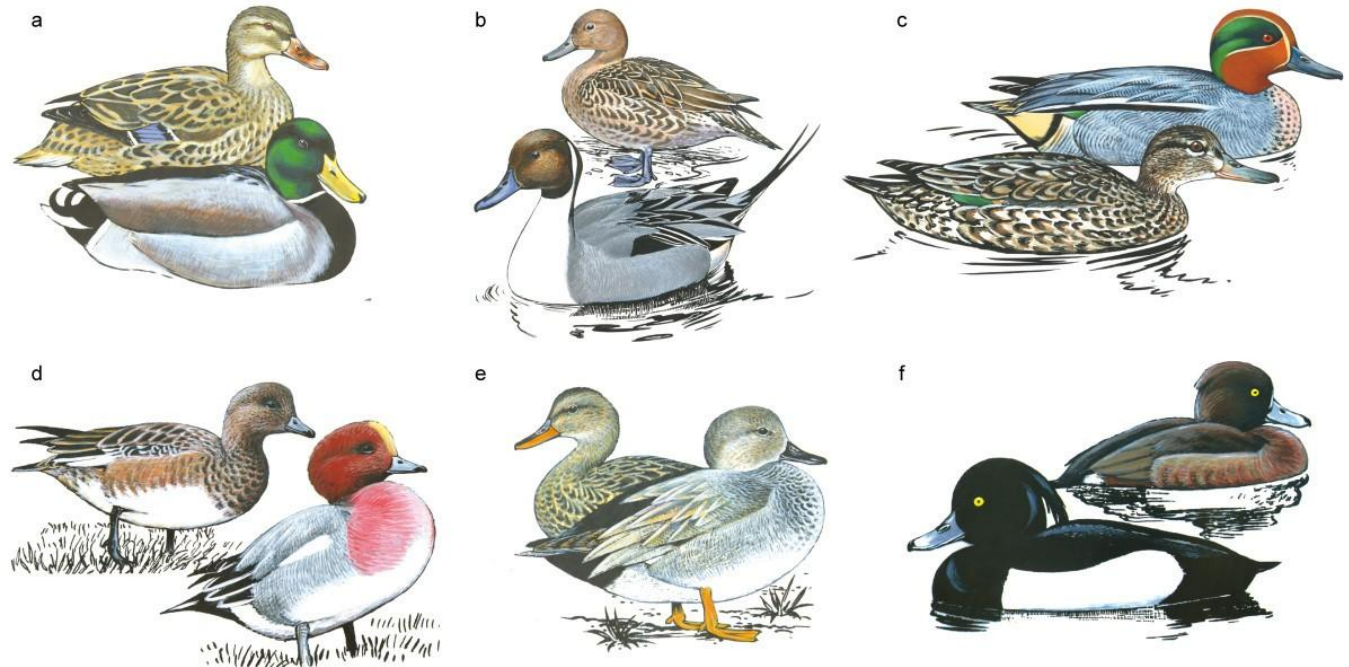


RESEARCH ARTICLE

Open Access

Widespread horizontal genomic exchange does not erode species barriers among sympatric ducks

Robert HS Kraus^{1,2*}, Hindrik HD Kerstens³, Pim van Hooft¹, Hendrik-Jan Megens⁴, Johan Elberg⁵, Arseny Tsvey⁶, Dmitry Sartakov⁷, Sergej A Soloviev⁸, Richard PMA Crooijmans⁴, Martien AM Groenen⁴, Ronald C Ydenberg^{1,9} and Herbert HT Prins¹



2. The challenge of Managing Hybrids in the Wild

The International Union for Conservation of Nature (IUCN) Red Listing process generally considers hybrids as a threat to biodiversity conservation (IUCN 2021).

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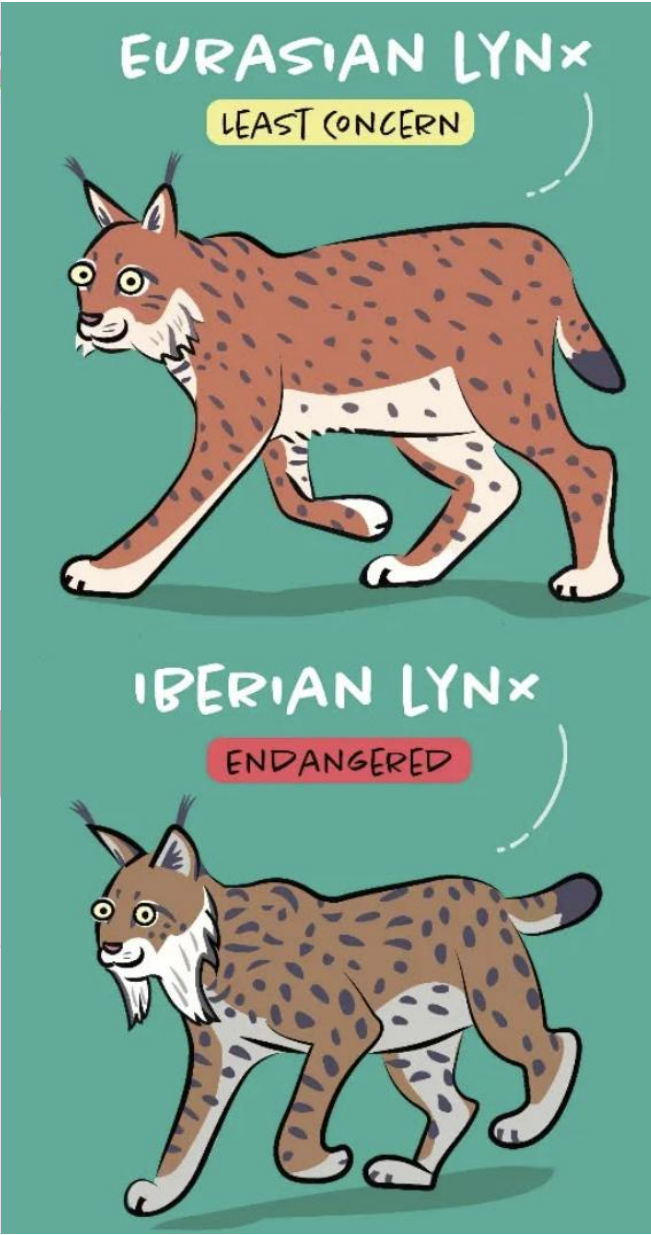
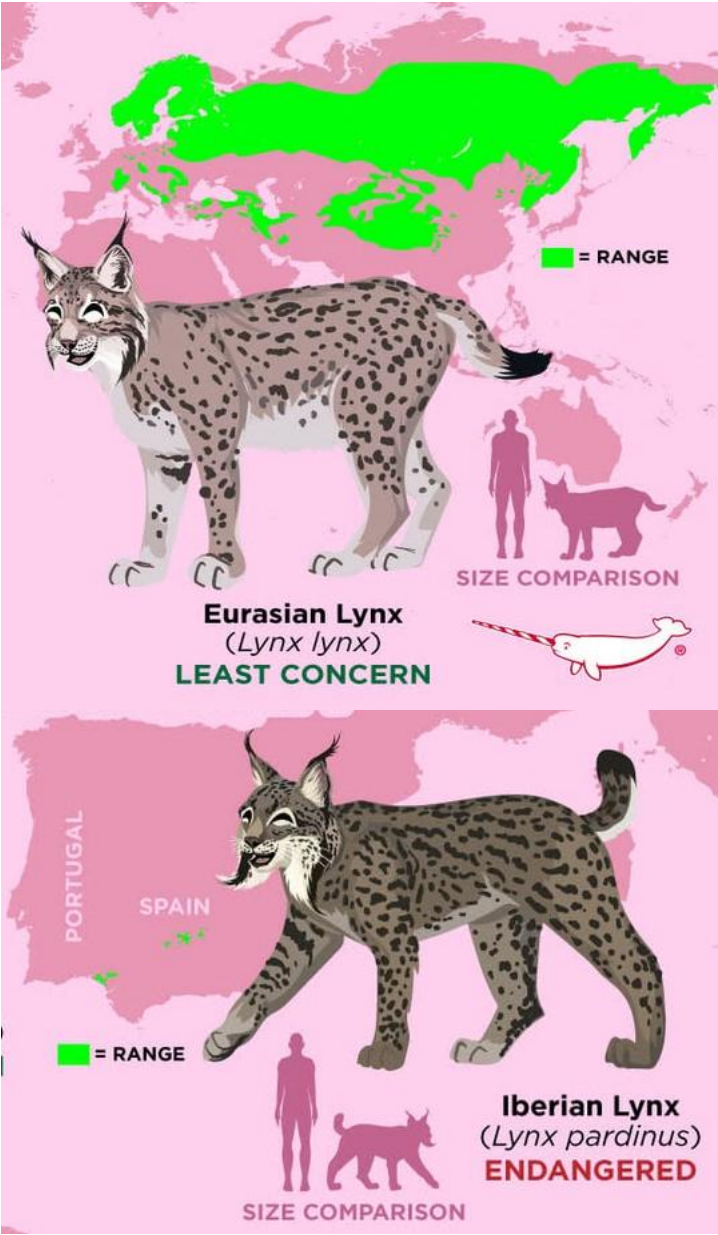
Why can this be a problem?

Ignoring hybrids can sometimes undermine conservation by restricting genetic diversity, which is linked to ecosystem resilience and long-term population survival. Cases where populations persisting through climate change had higher genomic diversity than those that went extinct.

Recent increase in species-wide diversity after interspecies introgression in the highly endangered Iberian lynx

Received: 30 June 2022
Accepted: 10 November 2023
Published online: 15 January 2024

Maria Lucena-Perez¹, Johanna L. A. Pajmans^{2,3}, Francisco Nocete⁴, Jordi Nadal⁵, Cleia Detry⁶, Love Dalén^{7,8}, Michael Hofreiter², Axel Barlow⁹ & José A. Godoy¹✉

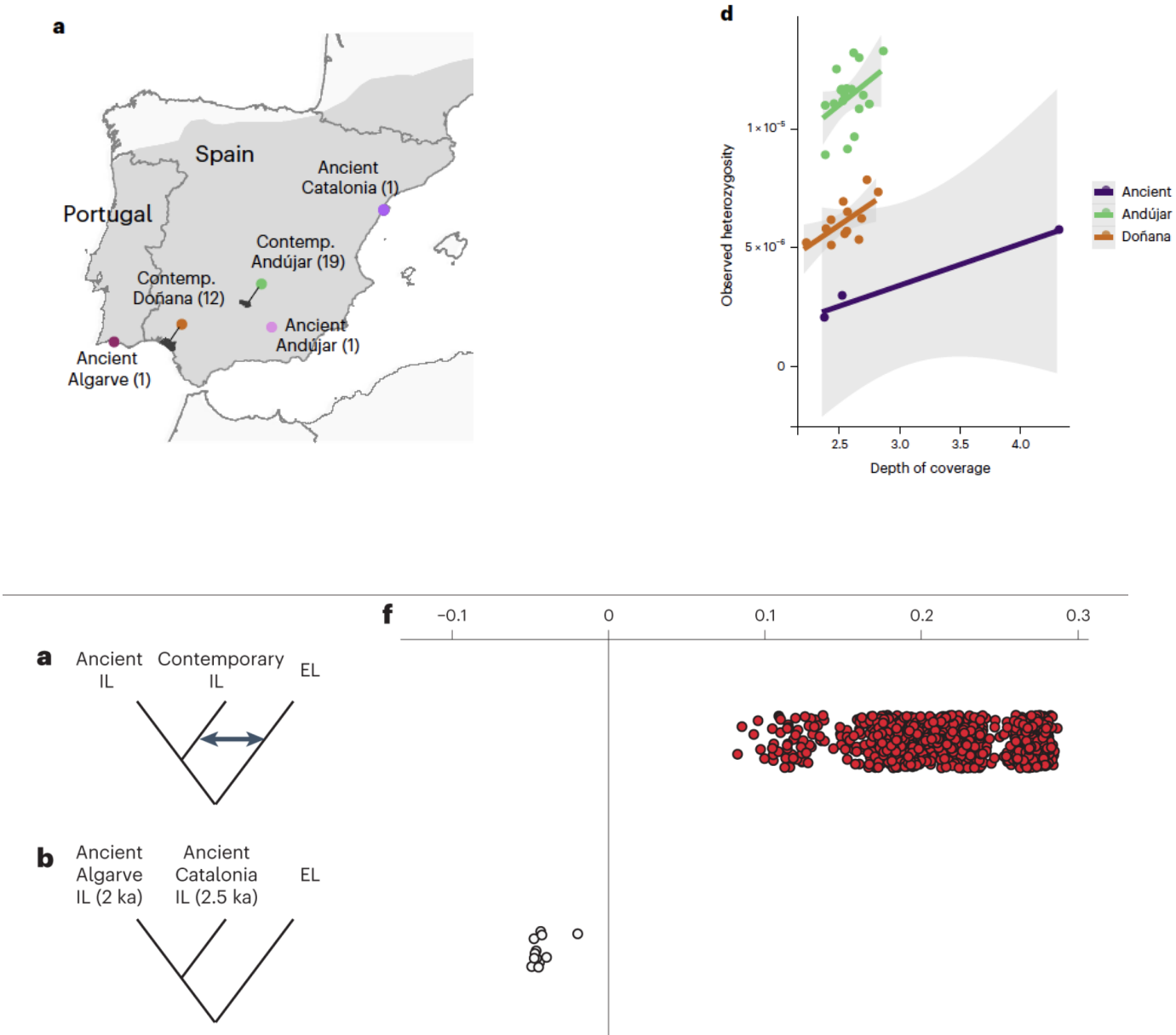


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- Ancient lynx had lower nuclear diversity than modern ones, suggesting later diversity gains via introgression.
- Gene flow was mainly Eurasian → Iberian
- Takeaway: introgression can help restore diversity in severely inbred species, raising questions for conservation policy and genetic rescue.



2. The challenge of Managing Hybrids in the Wild

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Growing emphasis on genetic diversity:

Conservation policy is increasingly encouraged to maintain genetic diversity for the future, aligned with international commitments such as the Kunming–Montreal Global Biodiversity Framework (CBD COP15, 2022).

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A major challenge is to integrate traditional, species-focused conservation approaches with the current understanding of hybrids in conservation policy and management.

3 | Current Policies, Legislations, Regulations, and Practice

Hybrid management in practice can take on many forms, but often it is a question of whether to **ignore or remove hybrids** (e.g., see bontebok case study in [SI3](#)). However, removal of hybrids may, at times, undermine long-term restoration efforts, especially for taxa that naturally experience a high frequency of hybridization (vonHoldt et al. [2018](#)). **At the global scale, hybrids (and backcrosses up to the fourth generation) are protected by CITES and the EU Wildlife Trade Regulations when at least one of the two “parents” is (CITES) listed in Appendix I or II** (Fossati [2024](#)). Other international legal instruments (i.e., the Habitats Directive and the Bern Convention) also allow for the protection or removal of hybrids. However, **this approach is currently not followed by the IUCN**. As such, the IUCN, with the exception of apomictic hybrid plants, excludes interspecific hybrids from the Red List of Threatened Species (IUCN [2021](#)), which limits the ability to evaluate the status of hybrids (Bauer et al. [2021](#)).

Coordination among countries

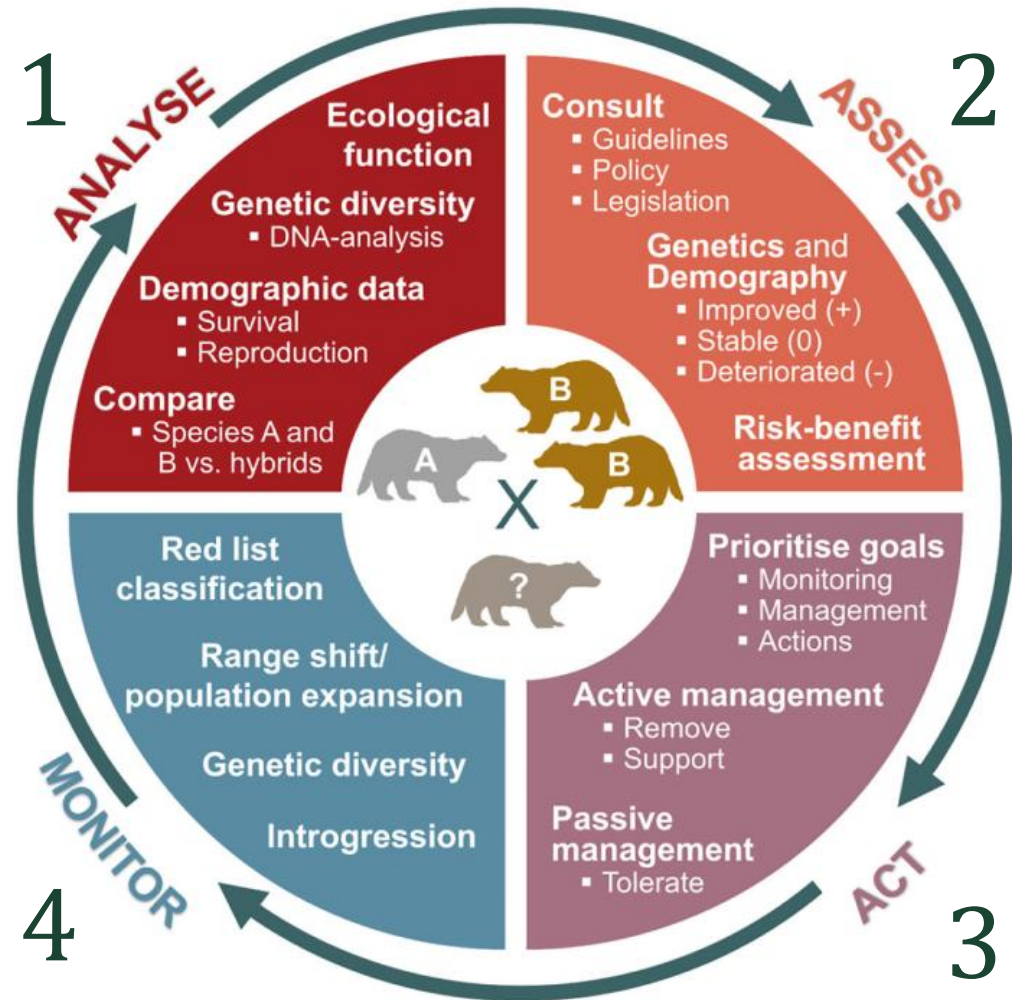
extant European wolf populations (Salvatori et al. 2020) In the absence of international policies on hybrid management, national interpretations of standing guidelines could lead to conflicting strategies even in neighboring countries (Salvatori et al. 2020), highlighting the need for clearer policies and more international coordination, including standardized genetic methods for the detection of hybrids. Trouwborst (2014) previously recommended for the Standing Committee of the Bern Convention and the European Commission to adopt a common and scientifically-based understanding of what hybrids are and to promote a uniform application of the legal instruments concerning hybrids.

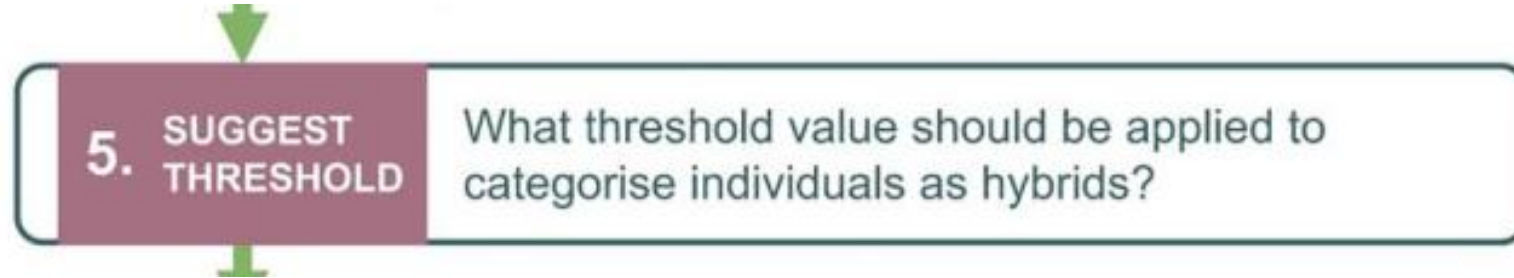
Table 2. Adoption of policies and management interventions to address wolf-dog hybridisation by countries that share the same wolf populations.

Population	Countries	Policy	Interventions
Alpine	Austria	Wolf management plan (2012) ^a	Removal
	France	Wolf Management plan (2018) ^b	Genetic monitoring
	Italy	No	Captivation
	Switzerland	Hunting ordinance Article 8 ^{bis} paragraph 5 (2012) ^c	Lethal removal

Proposed Framework for Future Conservation Policy and Management Including Hybrids

balance between
international harmonization
and
flexibility





that introgression-thresholds need to be decided (i) on a case-by-case basis, as part of the “Assess” step (Allendorf et al. 2001), (ii) in discussion with experts and stakeholders (using e.g., Population Viability Analysis simulations), (iii) with a clear understanding of the conservation objectives, and (iv) followed by regular review that allows for nuanced management (e.g., by integrating new information from high-resolution data and evolving taxonomies). Accordingly, it is important that scientists,

Main take-home message...

gap and the fact that they often lack legal status. Here, we recommend carefully evaluating hybrids with focus on the evolutionary consequences, including overall genetic diversity, adaptive variants, and ecological function, on a case-by-case basis.

Questions

? Is coordination between countries desirable, or is it enough for individual research groups to be able to communicate and collaborate with each other?

? Who should be responsible for “monitoring”: institutional authorities or universities?

? What evidence would justify action under uncertainty (and who decides)?

? Which outcomes matter most here: species identity, ecosystem function, or persistence?

? If hybridization is human-influenced, what is the “baseline” we aim for?

Thank you

Hybrids: threat, resource, or both?

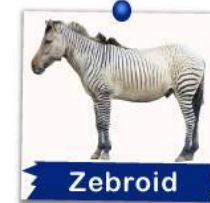
Hybrid Animals



Narluga



Geep



Zebroid



Dzo



Jaglion



coywolf



Zonkey



Cama



Pizzly



Toyger



Mulard



Liger



Wholphin



Savannah



Beefalo



Tigon



Zubron



Hinny



Leopon



Pumapard



Grolar Bear



Zorse



Mule